

ASSIGN4

September 2, 2026

```
[1]: import pandas as pd

# Enter your college roll number
roll_number = input("Enter your college roll number: ")

# Convert roll number to string
roll_number = str(roll_number)

# Extract last two digits
last_two_digits = roll_number[-2:]

print("Last two digits:", last_two_digits)
```

Enter your college roll number: 1024170414

Last two digits: 14

```
[2]: fixed_entries = [
    {
        "question": "what is the annual fee",
        "answer": "The annual fee is Rs 500.",
        "keywords": "fee cost price charge",
        "category": "billing"
    },
    {
        "question": "how to reset password",
        "answer": "Go to Settings > Reset Password.",
        "keywords": "password reset login",
        "category": "account"
    },
    {
        "question": "what are your working hours",
        "answer": "We are open 9 AM to 5 PM.",
        "keywords": "hours timing open time",
        "category": "general"
    },
    {
        "question": "how can i pay the fee",
```

```

        "answer": "You can pay via UPI, card, or net banking.",
        "keywords": "pay payment upi fee",
        "category": "billing"
    }
]

fixed_entries

```

```

[2]: [{ 'question': 'what is the annual fee',
        'answer': 'The annual fee is Rs 500.',
        'keywords': 'fee cost price charge',
        'category': 'billing'},
      { 'question': 'how to reset password',
        'answer': 'Go to Settings > Reset Password.',
        'keywords': 'password reset login',
        'category': 'account'},
      { 'question': 'what are your working hours',
        'answer': 'We are open 9 AM to 5 PM.',
        'keywords': 'hours timing open time',
        'category': 'general'},
      { 'question': 'how can i pay the fee',
        'answer': 'You can pay via UPI, card, or net banking.',
        'keywords': 'pay payment upi fee',
        'category': 'billing'}]

```

```

[3]: categories = ["billing", "account", "general"]

personalized_entries = []

for digit in last_two_digits:

    d = int(digit)

    category = categories[d % 3]

    if category == "billing":
        question = "how can i check my payment status"
        answer = "You can check your payment status from the billing section."
        keywords = "payment status billing transaction"

    elif category == "account":
        question = "how do i update my registered mobile number"
        answer = "You can update your registered mobile number from account_
↳ settings."
        keywords = "mobile number update account"

    else:

```

```

        question = "how can i contact customer support"
        answer = "You can contact customer support during working hours."
        keywords = "support help contact"

        personalized_entries.append({
            "question": question,
            "answer": answer,
            "keywords": keywords,
            "category": category
        })

personalized_entries

```

```

[3]: [{'question': 'how do i update my registered mobile number',
      'answer': 'You can update your registered mobile number from account
settings.',
      'keywords': 'mobile number update account',
      'category': 'account'},
      {'question': 'how do i update my registered mobile number',
      'answer': 'You can update your registered mobile number from account
settings.',
      'keywords': 'mobile number update account',
      'category': 'account'}]

```

```

[4]: all_entries = fixed_entries + personalized_entries

faq_df = pd.DataFrame(all_entries)

print("Final 6-row FAQ DataFrame:")
print(faq_df)

```

Final 6-row FAQ DataFrame:

	question \	answer \
0	what is the annual fee	The annual fee is Rs 500.
1	how to reset password	Go to Settings > Reset Password.
2	what are your working hours	We are open 9 AM to 5 PM.
3	how can i pay the fee	You can pay via UPI, card, or net banking.
4	how do i update my registered mobile number	You can update your registered mobile number f...
5	how do i update my registered mobile number	You can update your registered mobile number f...

	keywords	category
0	fee cost price charge	billing
1	password reset login	account
2	hours timing open time	general
3	pay payment upi fee	billing
4	mobile number update account	account
5	mobile number update account	account

```
[5]: print("Number of FAQ entries:", len(faq_df))
```

Number of FAQ entries: 6

```
[6]: def score_query(query, df):

    # Convert query into lowercase words
    query_words = set(query.lower().split())

    results = []

    for index, row in df.iterrows():

        # Combine question and keywords
        text = row["question"].lower() + " " + row["keywords"].lower()

        # Convert FAQ text into words
        faq_words = set(text.split())

        # Find matching words
        matched_words = query_words.intersection(faq_words)

        # Calculate score
        score = len(matched_words)

        if score > 0:

            # Confidence score
            confidence = score / len(query_words)

            results.append({
                "index": index,
                "question": row["question"],
                "answer": row["answer"],
                "category": row["category"],
                "matched_keywords": ", ".join(matched_words),
                "score": score,
                "confidence": round(confidence, 2)
```

```

    })

    # Sort by score from highest to lowest
    results = sorted(
        results,
        key=lambda x: x["score"],
        reverse=True
    )

    return pd.DataFrame(results)

```

```

[7]: query = input("Enter your question: ")

result = score_query(query, faq_df)

print("\nMatching FAQ entries:")
print(result)

```

Enter your question: 5

Matching FAQ entries:
Empty DataFrame
Columns: []
Index: []

```

[8]: def same_category(category_name, df):

    result = df[df["category"].str.lower() == category_name.lower()]

    return result

```

```

[9]: personalized_category = personalized_entries[0]["category"]

print("Personalized entry category:", personalized_category)

print("\nFAQs belonging to this category:")

print(same_category(personalized_category, faq_df))

```

Personalized entry category: account

FAQs belonging to this category:

	question \
1	how to reset password
4	how do i update my registered mobile number
5	how do i update my registered mobile number

```

                                answer \
1          Go to Settings > Reset Password.
4 You can update your registered mobile number f...
5 You can update your registered mobile number f...

```

```

                                keywords category
1          password reset login  account
4 mobile number update account  account
5 mobile number update account  account

```

```

[10]: # Display all FAQ entries
print(faq_df[["question", "keywords", "category"]])

# Select an entry
entry_number = int(input("\nEnter the row number (0 to 5) where you want to add_
↳ a keyword: "))

# Ask for new keyword
new_keyword = input("Enter a new keyword: ")

# Add the keyword
faq_df.loc[entry_number, "keywords"] = (
    faq_df.loc[entry_number, "keywords"] + " " + new_keyword
)

print("\nUpdated FAQ entry:")
print(faq_df.loc[entry_number])

```

```

                                question                                keywords \
0          what is the annual fee          fee cost price charge
1          how to reset password          password reset login
2          what are your working hours          hours timing open time
3          how can i pay the fee          pay payment upi fee
4 how do i update my registered mobile number  mobile number update account
5 how do i update my registered mobile number  mobile number update account

category
0 billing
1 account
2 general
3 billing
4 account
5 account

```

```

Enter the row number (0 to 5) where you want to add a keyword: 4
Enter a new keyword: discount

```

Updated FAQ entry:

```
question      how do i update my registered mobile number
answer        You can update your registered mobile number f...
keywords      mobile number update account discount
category      account
Name: 4, dtype: object
```

```
[11]: filename = roll_number + "_faq_data.csv"

faq_df.to_csv(filename, index=False)

print("Updated FAQ DataFrame saved successfully as:", filename)
```

Updated FAQ DataFrame saved successfully as: 1024170414_faq_data.csv

```
[12]: category_count = faq_df.groupby("category").size()

print("Number of FAQ entries per category:")
print(category_count)
```

Number of FAQ entries per category:

```
category
account    3
billing    2
general    1
dtype: int64
```

```
[13]: category_count = faq_df.groupby("category").size().reset_index(name="count")

print(category_count)
```

```
category  count
0  account      3
1  billing      2
2  general      1
```

```
[14]: def score_query_with_ties(query, df):

    # Convert query into lowercase words
    query_words = set(query.lower().split())

    results = []

    for index, row in df.iterrows():

        # Combine question and keywords
        text = row["question"].lower() + " " + row["keywords"].lower()
```

```

    # Convert FAQ text into words
    faq_words = set(text.split())

    # Find matching words
    matched_words = query_words.intersection(faq_words)

    # Calculate score
    score = len(matched_words)

    if score > 0:

        confidence = score / len(query_words)

        results.append({
            "index": index,
            "question": row["question"],
            "answer": row["answer"],
            "category": row["category"],
            "matched_keywords": ", ".join(matched_words),
            "score": score,
            "confidence": round(confidence, 2)
        })

    # If nothing matches
    if len(results) == 0:
        print("No matching FAQ found.")
        return pd.DataFrame()

    # Convert to DataFrame
    results_df = pd.DataFrame(results)

    # Find highest score
    highest_score = results_df["score"].max()

    # Select ALL entries having highest score
    best_matches = results_df[
        results_df["score"] == highest_score
    ]

    print("Highest Score:", highest_score)
    print("\nBest Matching FAQ(s):")
    print(best_matches)

    return best_matches

```

```
[15]: print("===== TIE EXAMPLE =====")
```

```
query = "fee"

score_query_with_ties(query, faq_df)
```

===== TIE EXAMPLE =====

Highest Score: 1

Best Matching FAQ(s):

	index	question	answer \
0	0	what is the annual fee	The annual fee is Rs 500.
1	3	how can i pay the fee	You can pay via UPI, card, or net banking.

	category	matched_keywords	score	confidence
0	billing	fee	1	1.0
1	billing	fee	1	1.0

```
[15]: index          question          answer \
0      0  what is the annual fee      The annual fee is Rs 500.
1      3  how can i pay the fee      You can pay via UPI, card, or net banking.
```

	category	matched_keywords	score	confidence
0	billing	fee	1	1.0
1	billing	fee	1	1.0

```
[16]: print("===== NO-TIE EXAMPLE =====")
```

```
query = "password"

score_query_with_ties(query, faq_df)
```

===== NO-TIE EXAMPLE =====

Highest Score: 1

Best Matching FAQ(s):

	index	question	answer category \
0	1	how to reset password	Go to Settings > Reset Password. account

	matched_keywords	score	confidence
0	password	1	1.0

```
[16]: index          question          answer category \
0      1  how to reset password      Go to Settings > Reset Password. account
```

	matched_keywords	score	confidence
0	password	1	1.0

```
[ ]:
```